

HDOC - Day 25

Menu-driven Decreasing Patterns

Algorithm

1. Start and display the menu containing the four decreasing patterns and Exit.
2. Read the user choice and use switch-case to call the selected parameterless function.
3. Read n inside the selected function.
4. Use nested loops with the row size decreasing from n to 1 to print the required pattern.
5. Return to the menu after each operation and stop when Exit is selected.

Test Case Table

Test	Operation	Input	Expected output
1	Decreasing star triangle	n = 4	4, 3, 2, 1 stars
2	Decreasing natural numbers	n = 4	1..4, 1..3, 1..2, 1
3	Repeated-number pattern	n = 4	4444, 333, 22, 1
4	Reverse-number triangle	n = 4	4 3 2 1 down to 1

C Program

```
#include <stdio.h>
void stars(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=n;i>=1;i--){for(j=1;j<=i;j++)printf("* ");printf("\n");}}
void natural(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=n;i>=1;i--){for(j=1;j<=i;j++)printf("%d ",j);printf("\n");}}
void repeated(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=n;i>=1;i--){for(j=1;j<=i;j++)printf("%d ",i);printf("\n");}}
void reverse(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=n;i>=1;i--){for(j=i;j>=1;j--)printf("%d ",j);printf("\n");}}
int main(){int ch;do{printf("\n1. Decreasing star triangle\n2. Decreasing natural-number pattern\n3. Decreasing repeated-number pattern\n4. Reverse-number triangle\n5. Exit\n");printf("Enter choice: ");scanf("%d",&ch);switch(ch){case 1:stars();break;case 2:natural();break;case 3:repeated();break;case 4:reverse();break;case 5:break;default:printf("Invalid choice\n");}}while(ch!=5);}
```

Execution

Test Case 1 - Decreasing Right-angled Triangle

```
puppie@Anvikas-MacBook-Air Desktop % clang hdoc25.c -o hdoc25
puppie@Anvikas-MacBook-Air Desktop % ./hdoc25
Enter choice: 1
Enter n: 4
* * * *
* * *
* *
*
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 2 - Decreasing Natural-number Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc25
Enter choice: 2
Enter n: 4
1 2 3 4
1 2 3
1 2
1
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 3 - Decreasing Repeated-number Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc25
Enter choice: 3
Enter n: 4
4 4 4 4
3 3 3
2 2
1
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 4 - Reverse-number Triangle

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc25
Enter choice: 4
Enter n: 4
4 3 2 1
3 2 1
2 1
1
puppie@Anvikas-MacBook-Air Desktop %
```

Result: The menu-driven C program using switch-case and nested loops was implemented and tested successfully for all four decreasing patterns.